

Two Gates, One Quantity: When Sampling Error Can Be Removed from a Conformal Calibration Set

Abstract

Repeated surveys produce estimated distributions for many populations, and those estimates are increasingly used to calibrate a prediction band for a further population. The calibration objects then carry sampling error of their own, so a band calibrated on them covers a further survey estimate but not automatically the underlying population distribution. We separate these two targets and ask when the sampling error can be removed. A conformal band treating populations as exchangeable gives finite-sample coverage for a further survey estimate; coverage for the underlying distribution requires either a sampling-error bound, which is shape-free and wide, or a scale correction, which requires the standardised observed and underlying curves to share a law. We then show that the two conditions an adaptive implementation of that correction gates on are not independent. The correction shrinks the target scale in proportion to the design share, and its reliability diagnostic is evaluated on that shrunken scale, so the requirement is a closed-form boundary in which the needed population count rises as the fourth power of the design share's complement. The implementation's reported floor of 94 populations is this boundary at a design share of zero, the regime where nothing needs removing; at the share where its own need gate opens the requirement is 153. Across 30 configurations of European Social Survey regional estimates the boundary predicts the gate correctly in all of them, while the fixed floor is cleared in 27 and the correction activates in 4. National estimates sit in the unnecessary regime. A simulation with exactly correct variances

shows that mismatched correlation structures cost simultaneous coverage in the direction expected for cumulative distributions, growing with the number of coordinates rather than shrinking with the number of populations.

Statement of Significance

A survey estimate is an uncertain description of a population. Using estimates from many populations to predict another introduces a second source of uncertainty, the differences between populations, and the two answer different questions. Methods exist that subtract estimated sampling variance to sharpen such a prediction. This paper shows that the conditions for using them are coupled rather than separate: the more sampling error there is to remove, the harder it is to certify what remains, because the reliability check is evaluated on the scale the correction shrinks. We give the resulting requirement in closed form and verify it on European survey data, where it predicts the method's own decision in every configuration examined while the fixed sample-size rule it replaces does not.

1 Introduction

Repeated probability surveys describe a population by a distribution rather than a mean. For an ordinal item the cumulative distribution function records the share of the population below each response threshold, and a survey programme covering many countries or many regions produces one such curve per population per round. These curves are increasingly used as inputs to a second analysis: information from the observed populations is used to construct a prediction band for a further one.

The inputs to that second analysis are estimates. Each carries sampling error from its own complex design, and a band calibrated on the discrepancies between those estimates and a prediction centre inherits that error. This paper asks what such a band guarantees, about which object, and under which assumption.

Two targets. Write $\tilde{F}_c$ for population c 's survey estimate and F_c for the population distribution behind it. A band may be asked to cover a further *survey estimate* $\tilde{F}_{K+1}$, which we call target T1, or the *population distribution* F_{K+1} , target T2. The two are not interchangeable. Exchangeability of the observed curves delivers T1 in finite samples by the ordinary conformal rank argument. It delivers nothing about T2 on its own, because the calibration scores contain sampling error that the latent target does not.

Reaching T2 requires additional information, and there are two routes. A bound on the target's own sampling error gives a shape-free enlargement of the T1 band. Alternatively, subtracting an estimate of sampling variance from the calibration spread gives a narrower band, at the price of a distributional assumption: that the standardised observed and latent curves follow a common law. Correct coordinate variances do not imply it.

What this paper adds. We do not claim novelty for the conformal rank argument, for the observation that measurement error is not identified from a convolution, or for simultaneous scoring of a curve. The contribution is to connect these to survey-estimated distribution curves and to characterise when the variance-subtraction route is reachable in practice.

The characterisation turns on a result we did not anticipate. An adaptive implementation of this correction gates on two quantities: whether there is enough design noise to be worth removing, and whether the remaining scale is determined precisely enough to trust. These read as independent requirements, and the implementation's own documentation treats them as two dimensions of a feasibility region. They are not independent. The deconvolution shrinks the target scale in proportion to the design share, and the reliability diagnostic is inflated by exactly that shrinkage. Across our survey applications the correlation between the design share and the diagnostic's inflation above its sample-size floor is 0.964. **Design noise large enough to be worth removing is, by that very magnitude, large enough to leave the remainder poorly determined.** The feasible

region is a narrow diagonal band, not a corner.

This displaces the account the implementation carried. Its documented obstacle was a population-count floor of ninety-four, derived from the diagnostic’s sampling term. In our regional applications that floor is cleared in 27 of 30 configurations, at population counts up to 290, and the correction still activates in only 4. The floor is not what binds.

Empirical characterisation. At the national unit of the European Social Survey the design share is 0.095–0.101 on full samples and reaches 0.369 in a subgroup scan; the need gate opens in none of 37 configurations. The correction is unnecessary there. One unit down, on regions within the same survey, the design share runs 0.49–0.66 and the correction becomes both needed and, in a few configurations, feasible. Which configurations depends on the variance estimator: a plain m -of- m resample opens the reliability gate in 8 of 8 arms and a Rao–Wu–Yue rescaled resample in 2 of 8, with the frozen threshold falling inside that gap.

Finally, a simulation isolates the distributional assumption from scale estimation. With exact coordinate variances throughout, mismatched correlation structures cost simultaneous coverage in a direction that is predictable for cumulative distribution functions, and the cost grows with the number of coordinates carried simultaneously rather than shrinking with the number of calibration populations.

2 Two prediction targets

Let $c = 1, \dots, K$ index calibration populations and $j = 1, \dots, d$ a fixed stack of response thresholds, and where applicable survey rounds. Relative to a prediction centre μ ,

$$Y_c(j) = \tilde{F}_c(j) - \mu(j) = G_c(j) + S_c(j), \quad G_c(j) = F_c(j) - \mu(j), \quad (1)$$

with S_c the survey estimation error and $v_c^2(j)$ its design variance. Conditional on a finite population, design-unbiased or approximately unbiased estimation motivates a centred error; for a weighted empirical distribution the usual ratio form is not exactly design-unbiased, and we keep the idealised centred model of the theory separate from its survey approximation.

The two statements of interest are

$$\Pr\{\tilde{F}_{K+1}(j) \in \hat{B}(j) \forall j\} \geq 1 - \alpha \quad (\text{T1}), \quad \Pr\{F_{K+1}(j) \in \hat{B}(j) \forall j\} \geq 1 - \alpha \quad (\text{T2}). \quad (2)$$

Both average over the population-level model and, for observed objects, their sampling mechanisms. Neither is conditional coverage for a named population. Exchangeability is assumed of the observed curves *together with their designs and observation patterns*; a shared label such as “country” does not establish it, and unlike designs can defeat it even when the latent curves are exchangeable. Departures are a separate problem with its own framework (Barber et al., 2023).

Table 1 states what each construction may claim.

Table 1: Constructions, the target each may claim, and the basis of the claim. All statements are simultaneous on a fixed finite grid.

Construction	Target	Basis
Observed anchor	T1	Exchangeable observed curves; finite-sample rank bound
Noise enlargement	T2	Anchor, widened by a bound on the target’s maximum sampling error
Oracle scale correction	T2	Common standardised-shape law (S), with known scales
Estimated correction	T2	(S), plus control of scale, centring and design-variance approximation error

Anchor. For a centre fixed independently of the curves, let $A_c = \max_j |Y_c(j)|$, $m = \lceil (1 - \alpha)(K + 1) \rceil$ and $q_\alpha = A_{(m)}$, with $q_\alpha = \infty$ when $m > K$. Exchangeability gives T1

coverage at least $m/(K+1)$. A radius that does not exist is reported as infinite rather than silently replaced by the sample maximum: the band is uninformative, and saying so is part of the guarantee.

Enlargement. If a deterministic $u \geq 0$ satisfies $\Pr\{\|S_{K+1}\|_\infty > u\} \leq \beta$, then a T1 band at level $1 - \alpha_0$ enlarged by u covers the latent curve with probability at least $1 - \alpha_0 - \beta$, by the triangle inequality on the intersection of the two events. No independence is needed and no shape is assumed. The route is shape-free but wide, and it is not operational for an unsurveyed population without information bounding the design error of its hypothetical survey.

Non-identification. Without information restricting S , the law of G is not identified by that of $Y = G + S$: distinct signal and noise laws generate the same convolution (Fan, 1991). This motivates the additional assumptions below. It does not establish that any particular conformal radius is optimal among latent-target procedures, and we make no width-optimality claim.

3 The scale correction and the assumption it needs

A narrower latent-target band is available under a stronger model. Suppose, for deterministic design scales,

$$Y_c(j) = \sqrt{s_G^2(j) + v_c^2(j)} W_c(j), \quad G_{K+1}(j) = s_G(j) W_{K+1}(j), \quad (3)$$

where $W_1, \dots, W_{K+1}$ are independent curves with a common law, zero coordinate means and unit coordinate variances. Call this assumption (S). The oracle scores $M_c = \max_j |Y_c(j)| / \sqrt{s_G^2(j) + v_c^2(j)}$ are then copies of $\|W\|_\infty$, and their conformal quantile placed on the latent scale gives a T2 band by the same rank argument.

Assumption (S) is stronger than a variance decomposition, and the difference is the subject of Section 5. Knowing $v_c^2(j)$ does not identify the cross-coordinate dependence of the latent scores. Gaussian latent curves and Gaussian sampling errors can violate (S) whenever their correlation matrices differ, because a standardised sum of independent processes need not share a summand's law. Substantively, (S) requires the sampling error to carry the same cross-threshold correlation profile as the population differences, so that noise changes only the scale of a common shape.

Write $s_Y^2(j) = s_G^2(j) + \overline{v^2}(j)$ for the observed spread and

$$\rho^2(j) = \overline{v^2}(j) / s_Y^2(j) \in [0, 1) \quad (4)$$

for the *design share*, the fraction of observed dispersion attributable to sampling. The oracle scale ratio is $s_G/s_Y = \sqrt{1 - \rho^2}$, so the correction's maximal proportional gain is $1 - \sqrt{1 - \rho^2}$, second order in ρ . That ratio is not the ratio of realised band widths, which also carries the ratio of two conformal quantiles.

Estimated scales. With estimated denominators differing from oracle values by relative error at most $\delta < 1$ outside an event of probability η ,

$$\Pr(\text{miss}) \leq \alpha + \eta + \Pr\{M_{K+1} \in (M_{(m)} \frac{1-\delta}{1+\delta}, M_{(m)}]\} , \quad (5)$$

which names the two further ingredients: control of scale-estimation error, and anti-concentration of the score law near the calibration quantile. Rates for these do not supply a numerical finite-sample level without their constants.

4 Two gates, one quantity

An adaptive implementation of the correction guards it with gates computed from the calibration set (Park, 2026). Two matter here. A *need* gate requires the design share to exceed a frozen cutoff, $\hat{\rho} > \rho_0$ with $\rho_0 = 0.47$. A *reliability* gate requires

$$D = \max_j \frac{\sqrt{2 s_Y^4(j)/(K-1) + (\text{SD}_K(\hat{v}^2(j))/\sqrt{K})^2}}{\hat{s}_G^2(j)} \leq \tau, \quad (6)$$

where $\hat{s}_G^2$ is the guarded scale, which subtracts a one-sided margin below the mean design variance so that an uncertain estimate subtracts less. At the frozen $\tau = 0.147$ the implementation reports the requirement $K \geq 94$, obtained by noting $\hat{s}_G^2 \leq s_Y^2$, whence $D \geq \sqrt{2/(K-1)}$.

That floor is correct and, as an account of what blocks the correction, misleading. The two gates are not independent requirements on two dimensions.

Proposition 1 (Gate coupling). *Let $\kappa(j) = \hat{s}_G^2(j)/s_Y^2(j) \in (0, 1]$ be the scale shrinkage and let*

$$h(j) = \frac{(\text{SD}_K(\hat{v}^2(j))/\sqrt{K})^2}{2s_Y^4(j)/(K-1) + (\text{SD}_K(\hat{v}^2(j))/\sqrt{K})^2}$$

be the share of the squared standard error contributed by dispersion of design variances across populations. Then at the coordinate attaining the maximum in (6),

$$D = \sqrt{\frac{2}{K-1}} \cdot \frac{1}{\kappa\sqrt{1-h}}, \quad (7)$$

and under (S), where the guarded subtraction recovers the design share, $\kappa \approx 1 - \rho^2$, so the reliability gate $D \leq \tau$ is

$$K \geq 1 + \frac{2}{\tau^2 (1 - \rho^2)^2 (1 - h)}. \quad (8)$$

Proof. Write $A = 2s_Y^4/(K-1)$ and $B = (\text{SD}_K(\hat{v}^2)/\sqrt{K})^2$ at the maximising coordinate, so $h = B/(A+B)$ and $A+B = A/(1-h)$. Then

$D = \sqrt{A+B}/\hat{s}_G^2 = \sqrt{A/(1-h)}/(\kappa s_Y^2) = \sqrt{2/(K-1)}/(\kappa\sqrt{1-h})$, which is (7).

Substituting $\kappa = 1 - \rho^2$ and solving $D \leq \tau$ for K gives (8). □

The content is the placement of ρ in (8). The deconvolution shrinks the target scale by $1 - \rho^2$, and the diagnostic divides by that shrunken scale, so the same design share that opens the need gate inflates the diagnostic by $(1 - \rho^2)^{-1}$ and works to close the reliability gate. **Design noise large enough to be worth removing is, by that magnitude, large enough to leave the remainder poorly determined.**

The reported floor of 94 is (8) evaluated at $\rho \rightarrow 0$ and $h \rightarrow 0$: it is the requirement in the regime where the correction is not needed. At the need cutoff itself, $\rho = \rho_0 = 0.47$, the requirement is already $K \geq 153$; at $\rho = 0.60$ it is 227, and at $\rho = 0.66$, the largest design share in our applications, 292.

Table 2: Populations required by (8) at $\tau = 0.147$ and $h = 0$. The reported floor of 94 is the entry at a design share of zero, where nothing needs removing. The need gate opens at $\rho_0 = 0.47$.

design share ρ	0	0.30	0.47	0.60	0.66
populations required	94	113	153	227	292

Section 6 checks (7) and (8) against 30 survey configurations. The identity holds to 3.2% mean and 7.1% maximum relative error, and the boundary predicts the gate correctly in 30 of 30 configurations, against 27 that the $K \geq 94$ rule would admit.

5 Isolating the shape assumption

Proposition 1 concerns whether the correction is *reachable*. Whether it is *sound* where reachable is a separate question, governed by (S), and coordinate variances do not answer it. We therefore hold the variances exactly correct and vary only the correlation shape.

Latent curves $G \sim N(0, R_G)$ and sampling errors $S \sim N(0, \sigma^2 R_S)$ are drawn independently with $\sigma = \rho/\sqrt{1 - \rho^2}$, so every coordinate variance is correct by construction and only R_G

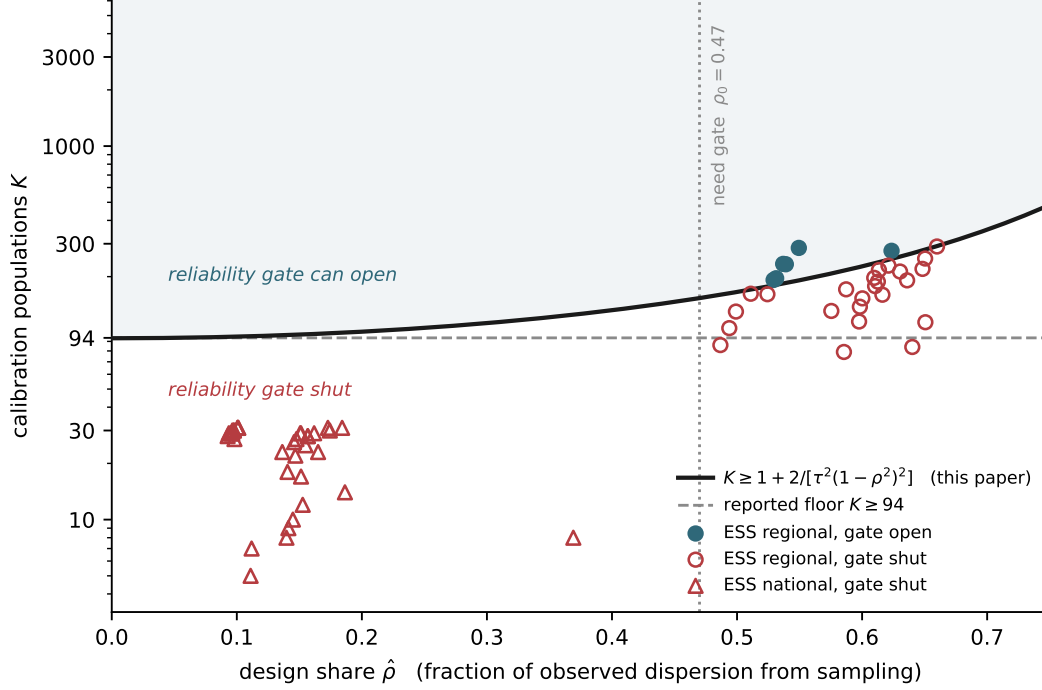


Figure 1: The feasibility boundary (8) at $\tau = 0.147$ and $h = 0$, with every survey configuration of Section 6. Filled markers are configurations where the reliability gate opened. The reported floor of 94 is the boundary’s intercept at zero design share; to the right of the need cutoff the requirement rises steeply, and the region between the two rules is where every configuration that clears the floor but fails the gate lies.

Alt text: Calibration population count on a logarithmic vertical axis against design share horizontally. A rising curve separates an upper region where the reliability gate can open from a lower region where it cannot. A horizontal dashed line at 94 marks the reported fixed floor, which coincides with the curve only at zero design share and falls increasingly far below it as the design share grows. Regional survey configurations cluster between design shares of 0.49 and 0.66 at counts near 100 to 290, straddling the curve; the four with the gate open lie above it. National configurations sit far below and to the left, at counts under 35 and design shares near 0.1.

versus R_S is manipulated. The grid crosses coordinate counts $d \in \{8, 24, 48\}$, design shares $\rho \in \{0.29, 0.40, 0.52, 0.80\}$, calibration counts $K \in \{30, 100, 250\}$ and five correlation pairs, at 4,000 replicates and nominal 90%: 180 cells. The first and third design shares are the maxima observed at the national and regional units in Section 6. These are standardised deviation vectors, not cumulative distributions and not a finite-population design, and they supply exact scales, so the experiment isolates (S) from scale-estimation error and cannot bound a deployed procedure’s realised coverage.

The direction is predictable, and the survey-realistic direction is the damaging one. For a cumulative distribution function, sampling errors at adjacent thresholds are partial sums over one sample and are strongly positively correlated; differences between populations across thresholds carry no such mechanism. The expected configuration is therefore noise more correlated than latent, and this was recorded as a falsifiable prediction before execution. It held: every structure with noise more correlated than latent undercovers, and the reverse overcovers, reaching 0.983 at $\rho = 0.80$, $d = 48$. A matched-correlation control sits at 0.900.

The cost grows with coordinates, not with populations. At $\rho = 0.80$ and $d = 48$ the worst mismatch gives latent coverage 0.755, against 0.859 at $d = 8$. Raising the calibration count does not repair it: the same cell gives 0.776 at $K = 30$ and 0.755 at $K = 250$. The mismatch is a property of the model, not a sampling error that more populations average away.

At the observed operating points the cost is small. At $\rho = 0.29$, the national maximum, no mismatch departs from nominal by more than Monte Carlo error (0.903). At $\rho = 0.52$, the regional maximum, the cost is one to two points (0.885). The severe failures require design shares above anything these surveys exhibit. The finding is therefore a warning about direction and mechanism, not a demonstration of failure in the applications

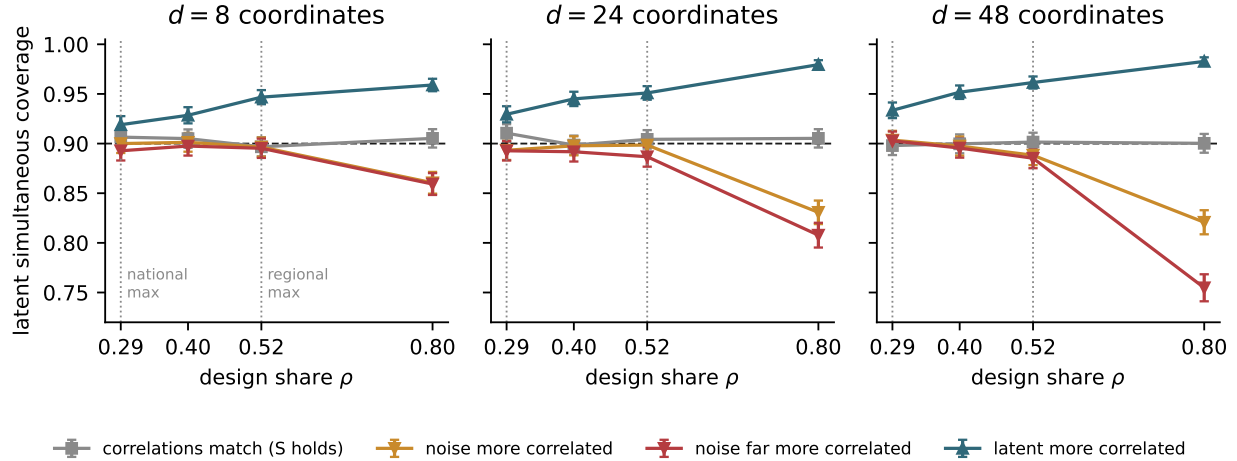


Figure 2: Latent simultaneous coverage of the oracle scale correction with exactly correct coordinate variances, by coordinate count, at $K = 250$ and nominal 90%. Bars are two Monte Carlo standard errors. Dotted verticals mark the largest design shares observed at the national and regional units.

Alt text: Three panels for eight, twenty-four and forty-eight coordinates. When the two correlation structures match, coverage stays at the nominal level throughout. When sampling error is more correlated across coordinates than population differences, coverage falls below nominal, mildly at the observed design shares and steeply at the highest one, and the fall deepens as the coordinate count rises. The reverse mismatch is conservative.

reported here, and we state it as such.

This is a post-development sensitivity analysis, designed after inspecting the assumption.

It is exploratory, not a preregistered confirmatory test, and it does not evaluate a deployed adaptive selector.

6 Survey applications

6.1 Data and design

We use European Social Survey rounds 9–11, which carry primary sampling unit, stratum, region and weight variables for all 33 participating countries (European Social Survey ERIC, 2024). Curves are weighted cumulative distributions of trust in parliament on the preregistered low-trust core $t \in \{1, 2, 3, 4\}$, the grid of the implementation under study; since D in (6) is a maximum over coordinates, the grid is not a cosmetic choice. Design

variances come from a Rao–Wu–Yue rescaled stratified bootstrap with $B = 400$ (Rao et al., 1992; Wolter, 2007).

The resampling unit. The regional estimand is a region’s departure from its own national distribution, and region and national curves are recomputed inside each replicate before differencing so that their covariance propagates. The resampling unit is (stratum, psu), never (stratum, psu, region): a primary sampling unit spanning two regions must enter a replicate whole, or the dependence the resample exists to represent is destroyed.

Whether this matters is an empirical question about the design files, and we audited all 90 country-rounds and 159,320 respondents. Sampling units nest inside regions for all but 0.52% of respondents, with violations in 15 of 65 country-rounds carrying informative units. Strata fail to nest in regions in 63 of 90. In 25 country-rounds the unit identifier is degenerate, equalling the respondent identifier: correct for individual-level register samples, but it means the pooled analysis mixes design variances that do and do not contain a clustering component, which we report separately rather than average over.

6.2 National estimates: the correction is unnecessary

At the national unit the design share is 0.095–0.101 on full samples, per round and for the stacked trajectory, and reaches 0.369 in a 33-cell subgroup scan by age band and minimum sample size. The need gate opens in 0 of 37 configurations. Whatever else is true, there is little sampling variance to remove relative to the dispersion between countries.

Consistently with Proposition 1, the dispersion term h is negligible here, 0.0–0.2%: at $K \leq 33$ the diagnostic is essentially its sample-size floor.

6.3 Regional estimates: needed, and mostly unreachable

One unit down the picture reverses. Across 30 configurations, two pools by three rounds by five minimum region sizes, the design share runs 0.49–0.66 and the need gate opens in 22.

The reliability gate opens in 6, and the correction activates in 4.

The sample-size floor is not the obstacle. It is cleared in 27 of 30 configurations, at counts up to 290. What binds is the coupling. The identity (7) reproduces the realised diagnostic to 3.2% mean and 7.1% maximum relative error, the correlation between the design share and the diagnostic’s inflation above its floor is 0.964, and the closed-form boundary (8) predicts the gate in 30 of 30 configurations against 27 admitted by the $K \geq 94$ rule. The four activations sit where the boundary says they can: at the smallest region size, and three of four in round 11, the round with the lowest design share.

6.4 Verification

Three checks accompany the above. The conclusion does not move with the bootstrap draw (four seeds, D within 0.001) or the replicate count ($B = 200$ to 2000, D within 0.002).

The third check separates the two departures from the inherited analysis. Under the archive’s own settings our implementation reproduces its reported diagnostic to four decimal places, so the comparison is between the same statistic computed two ways. The variance estimator is then decisive and the resampling unit is not: the reliability gate opens in 8 of 8 m -of- m arms and 2 of 8 Rao–Wu–Yue arms, with $\tau = 0.147$ falling inside that gap, while the resampling unit moves D by under 0.1% and never flips a gate. The latter follows from the design audit: with 0.52% of respondents in a region-splitting unit, there was little for that correction to move. A downward-biased variance estimator makes a correction look reachable that a defensible one does not.

7 Discussion

Prediction from survey-estimated curves requires naming the target. Exchangeability of observed curves yields a finite-sample guarantee for a further survey estimate, and nothing more. Reaching the population distribution requires either a sampling-error bound, which is shape-free and wide, or a scale correction, which is efficient and rests on a distributional assumption that coordinate variances cannot verify.

Where the correction is reachable is governed by Proposition 1, and the answer is narrower than a population-count threshold suggests. The quantity that makes a correction worth attempting is the quantity that makes its result hard to certify, because the diagnostic is evaluated on the scale the correction itself shrinks. A feasibility region drawn as “enough populations, and enough noise” is a diagonal band, and the reported floor of 94 is its intercept at a design share of zero, where nothing needs removing.

The survey evidence is consistent throughout. National estimates sit in the unnecessary regime. Regional estimates within the same survey generate real sampling noise, and mostly fail the reliability requirement at the very design shares that make them candidates. Whether the few surviving configurations activate depends on the variance estimator, which is not a reassuring place for a feasibility result to rest.

What we do not claim. No universal width-optimality for any latent-target band; the non-identification argument motivates assumptions rather than certifying a radius. No demonstrated improvement over existing noisy-calibration or small-area methods, since no matched comparison was run. Activation of a selector is not evidence that (S) holds: the gates read scale precision, and a procedure that abstains has not thereby proved its approximation accurate anywhere else. Coverage measured against survey estimates does not validate coverage for latent curves, and where the latent object is unobserved, as in the regional application, that check is unavailable in principle rather than merely absent.

For practice. Three levels should be reported separately and never interchanged: a finite-sample level for the observed target, a latent-target level with the assumptions and error terms that support it, and an empirical coverage assessment under a stated simulation design. Alongside them belong the unit assumed exchangeable, the grid, and the branch a selector returned. Where a correction is contemplated, (8) gives the population count required at the design share actually present, which is a more useful planning quantity than a fixed floor.

Open problems. Informative latent-target guarantees under weaker shape restrictions than (S), since Section 5 shows the cost of violating it is directional and grows with simultaneity. A version of (8) for biased, shrinkage, or structured-covariance variance estimators, which the present statement excludes and which are exactly the estimators small-area practice is built on. And design-variance estimation that does not force a choice between a downward-biased scheme under which a correction appears reachable and a defensible one under which it does not.

Data and Code Availability

Code, protocols, a claims ledger binding every reported number to the artefact that produced it, and the outputs of every experiment are in the replication repository. The simulation of Section 5 requires no microdata. European Social Survey microdata are distributed under the provider’s access conditions and are not redistributed.

Analysis Planning

Each experiment has a protocol written and committed before execution, stating the outcomes that would count against the hypothesis. The sensitivity analysis of Section 5 was designed after methodological development and is exploratory; its full grid is reported.

One earlier version of the regional analysis reported a result that a verification run overturned, and the protocol now requires a reimplementation to reproduce an inherited statistic before any disagreement with it is reported.

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